

Student Performance Predictor

Overview

Student Performance Predictor is a beginner-friendly Machine Learning project developed using Python and Scikit-Learn. The project predicts whether a student is likely to **Pass** or **Fail** based on key academic factors such as attendance, study hours, and marks.

This project demonstrates the basic Machine Learning workflow, including data collection, model training, prediction, and evaluation.

Features

- Predicts student performance (Pass/Fail)
 - Uses a Machine Learning Decision Tree Classifier
 - Simple and easy-to-understand implementation
 - Beginner-friendly project for Data Science learners
 - Command-line based prediction system
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Technologies Used

- Python
 - Pandas
 - Scikit-Learn
 - Pickle
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Project Structure

```
student-performance-ml/  
|  
├─ data.csv  
├─ train.py  
├─ predict.py  
├─ model.pkl  
├─ requirements.txt  
└─ README.md
```

Dataset

The dataset contains the following features:

Feature	Description
attendance	Student attendance percentage
study_hours	Daily study hours
marks	Internal examination marks
result	Pass or Fail

Machine Learning Algorithm

The project uses a **Decision Tree Classifier** from Scikit-Learn to learn patterns from student data and make predictions.

Installation

1. Clone the Repository

```
git clone https://github.com/your-username/student-performance-ml.git
cd student-performance-ml
```

2. Install Dependencies

```
pip install -r requirements.txt
```

Training the Model

Run the following command:

```
python train.py
```

Output:

Model Trained Successfully!

A trained model file named `model.pkl` will be generated.

Making Predictions

Run:

```
python predict.py
```

Example:

```
Enter Attendance (%): 85
Enter Study Hours: 3
Enter Marks: 75

Prediction: Pass
```

Learning Outcomes

Through this project, I learned:

- Data preprocessing using Pandas
- Training Machine Learning models
- Decision Tree Classification
- Model serialization using Pickle
- Making predictions using trained models
- Basic Data Science workflow

Future Improvements

- Add a graphical user interface using Streamlit
 - Increase dataset size for better accuracy
 - Add visualization and analytics dashboards
 - Predict multiple performance categories instead of Pass/Fail
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Author

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